

Haoquan Wang

wanghaoquan0117@gmail.com | <https://www.linkedin.com/in/haoquanwang/>

EDUCATION

University of Utah

Bachelor of Science in Computer Science

Minor: Mathematics

Academic Awards:

Dean's List | Fall 2025 – Present

Utah Global Scholarship | Spring 2025 - Present

Salt Lake City, Utah

Jan 2025 - Expected May 2028

WORK

Wuhan Changhang Keda Engineering

Office Staff (40 hours per week)

Wuhan, Hubei, China

August 2022 - October 2023

- Audited complex hydrological datasets against national engineering standards, identifying 2 compliance discrepancies that prevented potential regulatory failures.
- Optimized systematic data cross-referencing procedures, increasing statistical accuracy by 3%.
- Evaluated engineering project performance by synthesizing multi-dimensional data trends, providing data-driven insights that mitigated compliance risks by 7%.

PROJECTS

Autonomous Path-Finding Drone | C++, Arm Mbed OS

July 2023 - March 2024

- Engineered a navigation drone using C++ on the Arm Mbed OS platform, implementing real-time sensor data processing for autonomous flight.
- Optimized control algorithms to bridge the gap between high-level path planning and low-level motor execution, significantly improving flight stability.

Data Structures & Algorithms Implementation | Java (GT CS 1332 Curriculum)

May 2025 - September 2025

- Developed 10+ core data structures from scratch, including External Chaining HashMaps, AVL Trees, and MaxHeaps, adhering to GT's strict $O(\log n)$ efficiency standards.
- Implemented complex graph algorithms such as Dijkstra's and Prim's, ensuring robust memory management and algorithmic optimization.

Deep Learning: Image Classification with Data Augmentation | Python, PyTorch, CUDA

August 2025 - October 2025

- Implemented a comprehensive CIFAR-10 image classification experiment comparing baseline and augmented training approaches, achieving 4% test accuracy improvement (87% → 91%) while effectively preventing overfitting.
- Engineered GPU-accelerated training pipeline using PyTorch with CUDA 11.8, optimizing batch processing and mixed-precision training to achieve 10-20x speedup over CPU implementation.
- Applied multiple data augmentation techniques (RandomHorizontalFlip, RandomCrop, ColorJitter, RandomRotation) combined with Dropout and BatchNormalization, reducing

overfitting gap from 13% to near-zero and significantly enhancing model generalization on unseen data.

Deep Learning: Function Fitting & Image Enhancement | Python, PyTorch

August 2025 - December 2025

- Constructed neural networks using PyTorch to perform high-precision function approximation and data regression tasks.
- Integrated data augmentation techniques to train lightweight models, successfully increasing classification accuracy on limited datasets.

Classic Snake Game Implementation | Python, Pygame

June

2025 - August 2025

- Developed a fully functional Snake game using the Pygame library, implementing core game mechanics including coordinate-based movement and collision detection.
- Designed an efficient event-handling loop to manage real-time user input and game state updates, ensuring smooth 60 FPS performance.
- Engineered a dynamic growth algorithm for the snake's body using List/Array data structures, maintaining consistent logic as game complexity increased.

SKILLS

Languages: Python, Java, C/C++

Tools: PyTorch, TorchVision, Numpy, Mbed OS, Git, JUnit, Pygame